135 PIN HEADER

J1A
A1/A19
A2/A20
GND
GND
12V
12V
A10/A1
A11/A2
A12/A3
A13/A4
A14/C12
A15/C13
A16/A14
12V
A18/B13
5V_ECU
A21/A27
A22/A28
A23/A6
A24/A8
A25/A5
A26/A7
A27/A15
A28/B14
AGND
A31/A29
A32/A30
A33/A25
12V
1-1376430-8

J1C
12V
GND
5V_ECU
AGND
135H_C12
135H_C13
135H_C15
C16/C8
CANHH_P
135H_C18/60H_B20
AGND
C20/B19
C22/B8
CANHH_N
135H_C29/C14
C30/A13
C31/B11
C33/A18
1-1376430-8

J1B
135H_B6/60H_C21
12V
135H_B9
B11/A32
B12/A31
135H_B13/60H_A34
135H_B15/60H_C23
135H_B17/60H_A21
GND
5V_ECU
B23/B15
B26/A22
AGND
B31/B16
135H_B32/60H_C24
135H_B33/C9
B35/A12
1-1376430-8

J1D
D2/B23
GND
GND
135H_D6/C5
12V
135H_D9/34H_B18
AGND
D10/B27
D11/B28
135H_D15/C6
D16/B3
D17/B1
D20/B10
D22/B12
D23/C11
D24/B5
D25/B2
D29/C10
AGND
1-1376430-8

60 PIN HEADER

J2A
A10/A1
A11/A2
A12/A3
A13/A4
A25/A5
A23/A6
A26/A7
A24/A8
60H_A9/60H_C22
GND
B35/A12
C30/A13
A16/A14
A27/A15
(A16)/C18
(A17)/C19
C33/A18
A1/A19
A2/A20
135H_B17/60H_A21
B26/A22
CANHH_P
CANHH_N
A33/A25
A21/A27
A22/A28
A31/A29
A32/A30
B12/A31
B11/A32
60H_A33/60H_C15
135H_B13/60H_A34
6437288-3

J2B
12V
GND
CAN1_P
CAN1_N
135H_D6/C5
135H_D15/C6
D-4/C7_#A2
C16/C8
135H_B33/C9
D29/C10
D23/C11
A14/C12
A15/C13
135H_C29/C14
60H_A33/60H_C15
(B6)/C16
(B7)/C17
(A16)/C18
(A17)/C19
34H_B17/ 60H_C20
135H_B6/60H_C21
60H_A9/60H_C22
135H_B15/60H_C23
135H_B32/60H_C24
5V_ECU
AGND
6437288-3

34 PIN HEADER

J3
D17/B1
D25/B2
D16/B3
D24/B5
(B6)/C16
(B7)/C17
C22/B8
D20/B10
C31/B11
D22/B12
A18/B13
A28/B14
B23/B15
B31/B16
34H_B17/ 60H_C20
135H_D9/34H_B18
C20/B19
135H_C18/60H_B20
CAN1_P
CAN1_N
D2/B23
5V_ECU
AGND
D10/B27
D11/B28
6437288-2

CAN RESISTOR

CANHH_P R5 120R CANHH_N

MCP2515_CAN

C1 10uF
C2 0.1uF
5V_REG
GND
U3
RESET# 19
RXCAN 2
CS# 18
SCK 14
MOSI 16
MISO 17
OSC1 9
OSC2 8
VDD 20
RESET#
RXCAN
CLKOUT/SOF 3
CS#
SCK
SI
SO
TX0RTS# 4
TX1RTS# 5
TX2RTS# 7
OSC1
OSC2
INT# 13
TXCAN 1
RX0BF# 12
RX1BF# 11
NC 6
NC 15
VSS 10
MCP2515T-I/ST
GND

C3 22PF
OSC1
8Mhz
X1
GND
C4 22PF
OSC2

U4
TXCAN 1
5V_REG 3
RXCAN 4
TXD
GND
VCC
RXD
S
CANH
CANL
VREF
8 R6 0R
7 CAN1_P
6 CAN1_N
5
GND
C14 0.1uF
GND GND
TJA1050T/CM,118

CAN1_N R7 120R CAN1_P
J4
826629-2
NOTE: Add jumper for male header in BOM

5V_REG R8 10K RESET#

BUCK 12V → 5V

12V
GND
C6 47uF
C7 47uF
C9 10uF
U7
VIN 3
EN 2
BST 6
SW 5
FB 1
GND 4
AP63205WU-7
0.1uF
C8
5V_REG
GND
L1 4.7uH
C10 22uF
C11 22uF
5V_REG
C12 47uF
C13 47uF
GND

Arduino Nano

GND
135H_C13_R2 R1 10K D1 135H_C13_A0
135H_C12_R3 R4 10K D2 135H_C12_A1
GND
D-4/C7_#A2 R10 R9 10K D3 D-4/C7_A2
GND
135H_C15_R11 R15 4.7K D4 135H_C15_D7
135H_B9_R12 R18 4.7K D5 135H_B9_D8
GND

U5
GND
INT# 5
135H_C15_D7 10
135H_B9_D8 11
CS# 13
MOSI 14
MISO 15
D0/TX 1
D1/RX 2
RESET_1 3
GND_1 4
D2 6
D3 7
D4 8
D5 9
D6 10
D7 11
D8 12
D9 13
D10 14
D11 15
D12 16
VIN 30
GND_2 29
RESET_2 28
5V 27
A7 26
A6 25
A5 24
A4 23
A3 22
A2 21
A1 20
A0 19
AREF 18
3V3 17
D13 16
10uF
C5 5V_REG
GND
D-4/C7_A2
135H_C12_A1
135H_C13_A0
SCK
A000005

eSpark Consulting Group

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